

Metric switching: tapping analysis

Roberto Barumerli

2026-07-17

Contents

What this document is	1
Part 1: One trial, up close	1
Is this trial typical?	2
Part 2: What the model does	3
Part 3: Spontaneous switches	5
A model-free check: does the accent pattern itself show a switch?	5
Aligning spontaneous trials on the model's estimated switch	6
A closer look: a small timing bump right after the switch	7
Part 4: Comparing conditions and meters	7
What about <code>sigma_0</code> ?	8
Does musical training predict switch behaviour?	9
Summary	10
What works, what's next	10

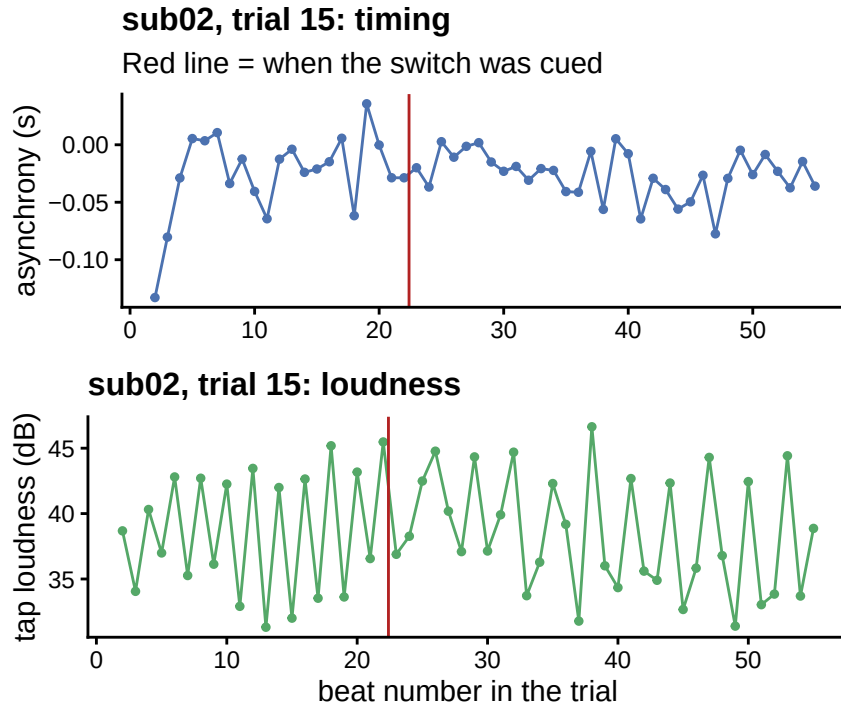
What this document is

Participants tapped along with a metronome. Partway through some trials, the underlying meter changed (a “switch”) – either **cued** (an external signal told them when) or **spontaneous** (no signal; if a switch happens, it's the participant's own doing). We recorded every tap's timing and loudness and want to know: how does tapping behaviour change around a switch, and can we estimate *when* a switch happened even when nobody told the participant to switch?

Based on 6 participants (sub02, sub04-08) with usable recordings; sub01 was excluded (too quiet/noisy to trust) and sub03 never had a trial log, so extraction couldn't run for it. Every trial is still fit completely independently, exactly as in the original single-subject version – what's new is pooling those independent per-trial results across participants for the group comparisons in Part 4, not a hierarchical/shared-parameter model (deliberately kept simple).

Part 1: One trial, up close

Before any pooling or modelling, here's a single cued trial in full: every tap's timing (top) and loudness (bottom), with the red line marking exactly when the participant was cued to switch meter.



Look at the two panels around the red line. In the **timing** panel, it's hard to tell anything happened at all – the tap-to-click offset looks about as noisy before as after. In the **loudness** panel, though, there's a visible change in *pattern*: before the switch, every other beat is noticeably louder (a clear accent, alternating strong-weak); after it, that particular alternation is gone.

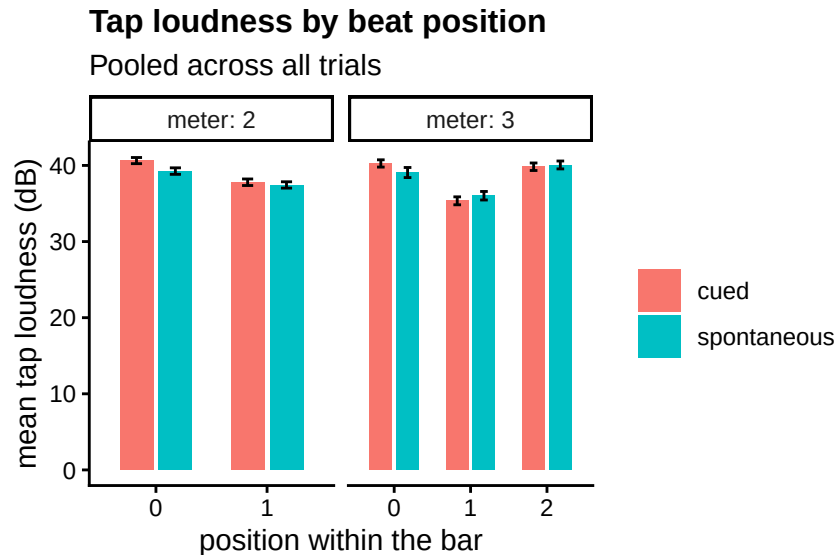
Is this trial typical?

Subject	Cued	Spontaneous
sub02	10	10
sub04	10	10
sub05	10	10
sub06	10	10
sub07	10	10
sub08	10	10

6 participants, 120 trials total, split evenly between cued (an instruction tells the participant when to switch) and spontaneous (no instruction) within each participant, and between the two meters tested (2 and 3 beats per bar). Tempo ranged 100-120 BPM across trials. Pooled across every trial, timing shows no obvious *switch-related* structure, confirming the single trial above wasn't a fluke – **timing alone doesn't show a reliable switch signal**.

Subject	Mean asynchrony (ms)	SD (ms)
sub02	-25.5	39.5
sub04	-53.1	41.0
sub05	-56.8	40.6
sub06	-45.5	34.7
sub07	-70.0	118.4
sub08	-39.6	25.3

It isn't exactly zero, though: every participant taps slightly *before* the click on average (all negative, roughly -25 to -70 ms) – a well-known anticipation effect in tapping tasks (negative mean asynchrony), not something specific to this experiment. It doesn't change the switch-detection story above since it's a constant offset, not a switch-related pattern.



Loudness, on the other hand, generalises: across trials and both meters there's a consistent loudness contrast between beat positions (a louder “downbeat”). **This is why the model below is built entirely on loudness, not timing** – it's the channel that actually carries the switch signal.

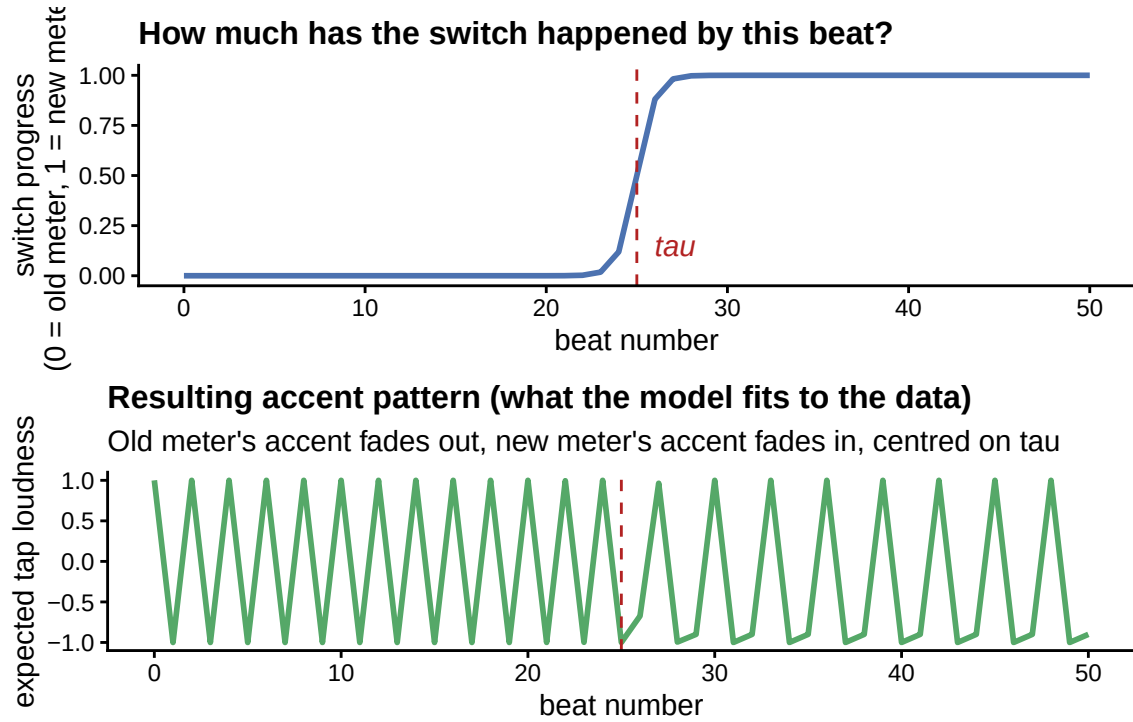
Part 2: What the model does

We fit a statistical model to each trial's tap-loudness pattern. In plain terms, the model looks for **the point in the trial where the participant's tapping pattern changes from one meter's accent to the other's**, and reports:

- **tau** – **when** the change happened, given as a beat number in the trial (or, for cued trials, we can compare it directly to when the instruction was given).
- **how confident** the model is in that estimate, from the *width* of **tau**'s credible interval – a trial with a clear, well-defined switch gives a narrow, tight estimate; a trial where the data don't show a clear switch (or don't show one clearly enough to pin down) gives a wide, uncertain estimate instead of a falsely precise one.

Concretely, each beat's loudness (in dB) is converted to a log-amplitude number, and that's the one value the model fits at every beat – expected to sit near one meter's accent pattern before the switch and the other's after it, blending smoothly in between. (Full formulas in `../docs/model_math.md`.)

The model allows the switch to be gradual, but doesn't assume it has to be – checking this directly against the data (comparing a smooth-transition version of the model against a version where the switch is instantaneous) shows real switches are close to instantaneous: tapping style changes over roughly a beat, not gradually across many beats. Either way, the model only uses the loudness/accent information described in Part 1 – not raw timing – to find the change point.

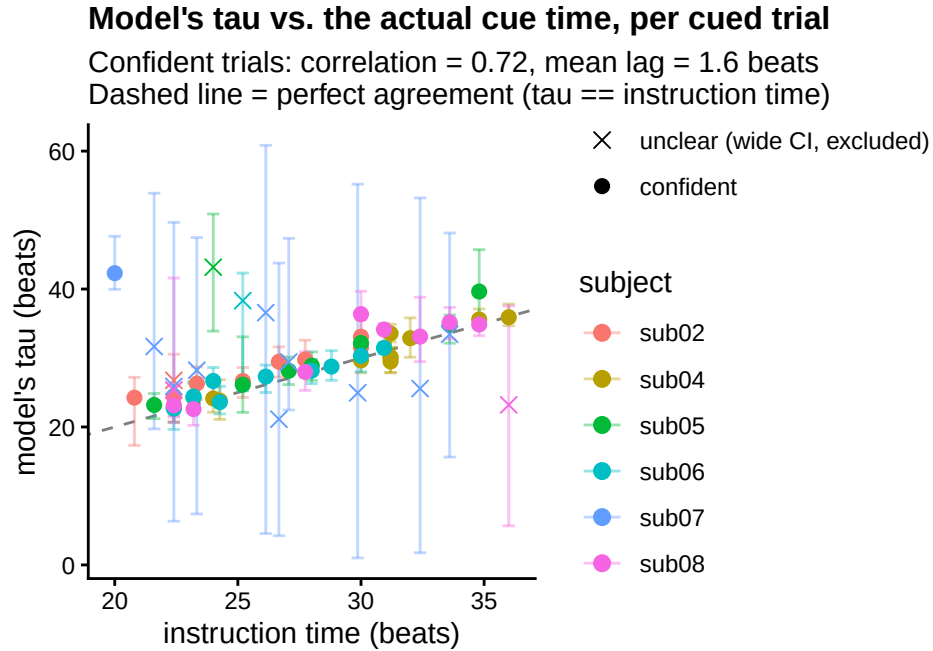


(*Illustrative example, not a fit to real data.*) Top: the model’s “switch progress” curve – a smooth S-shape that goes from 0 (still in the old meter) to 1 (fully in the new meter), centred on τ . Bottom: this progress curve blends the old meter’s accent pattern into the new one’s, which is what the model actually compares against the real tap-loudness data. The model can fit either a narrow (quick) or wide (gradual) S-curve, whichever the data support – and as noted above, real trials consistently favour a narrow one, close to an instant switch.

We do **not** detail the model’s internal mechanics beyond this here; see `../docs/model_math.md` and `02_fit_model.ipynb` for the full specification.

Validation. Before trusting the model on the harder spontaneous case, we checked it against the cued trials, where we know when the *instruction* to switch was given. τ isn’t meant to match that instruction time exactly – it’s fit purely from tap loudness, so it measures when the participant’s *tapping itself* actually changed, which should come a little after the instruction (reaction time, plus a moment to settle into the new pattern). Some cued trials have a genuinely wide, sometimes multi-modal τ posterior – checked directly (not a sampler bug: R-hat is fine, chains agree with each other on the same wide, lumpy distribution), this happens specifically when a trial’s accent contrast is weak, the same failure mode already seen for the spontaneous side. We apply the exact same fix here: flag trials with a wide 94% credible interval as “unclear” rather than trusting a point estimate the model itself is unsure about. Restricted to the confident trials, correlation with instruction time is 0.72 (vs. 0.45 unfiltered) and lag is far less noisy (-1.7 to +22.3 beats, vs. -12.8 to +22.3 unfiltered).

Subject	N trials	N confident	Mean lag (beats)
sub02	10	9	2.20
sub04	10	10	-0.07
sub05	10	9	1.59
sub06	10	9	0.62
sub07	10	1	22.30
sub08	10	8	1.53



Part 3: Spontaneous switches

With no cue, there's no obvious external marker to tell us when (or whether) a participant switched on their own. This is exactly where the model's **tau** estimate helps: for each trial, we can ask *when the model thinks the participant changed*, and *how sure it is* (from the width of **tau**'s credible interval).

Subject	N spontaneous trials	N confident switches
sub02	10	9
sub04	10	9
sub05	10	10
sub06	10	9
sub07	10	2
sub08	10	5

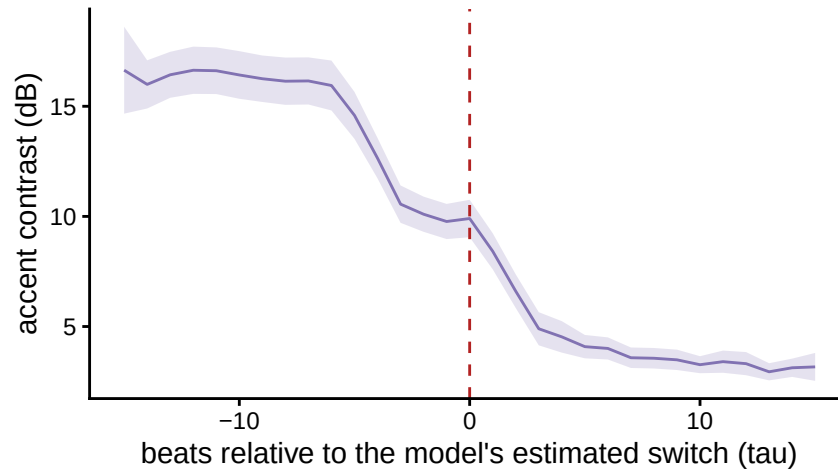
44 of 60 spontaneous trials show a confident, well-identified switch overall, but that varies a lot by participant (from 2/10 to 10/10) – some people's spontaneous switches are just harder to pin down than others'. The rest are ambiguous (a wide **tau** credible interval means the data don't clearly pin down when – or whether – the participant's tapping pattern actually changed). We only use the confident trials below – mixing in the ambiguous ones would add noise from switch estimates the model itself is unsure about.

A model-free check: does the accent pattern itself show a switch?

Before trusting the model's **tau** estimates, it's worth asking whether a switch is visible directly in the raw loudness data, with no fitting at all. For each trial, we slide an 8-beat window along the trial and compute one simple number: **how much loudness varies across the different positions in the bar, right now** (the gap between the loudest and quietest position, in dB, using only the taps in that window). If a participant has a clear accent pattern, this number is high; if it fades to a flat, unaccented tapping style, this number drops toward zero. A clear, sustained drop is a model-free signature of a switch.

Model-free accent contrast, aligned on the model's

Mean $\pm$ SE across the 44 confident-switch trials

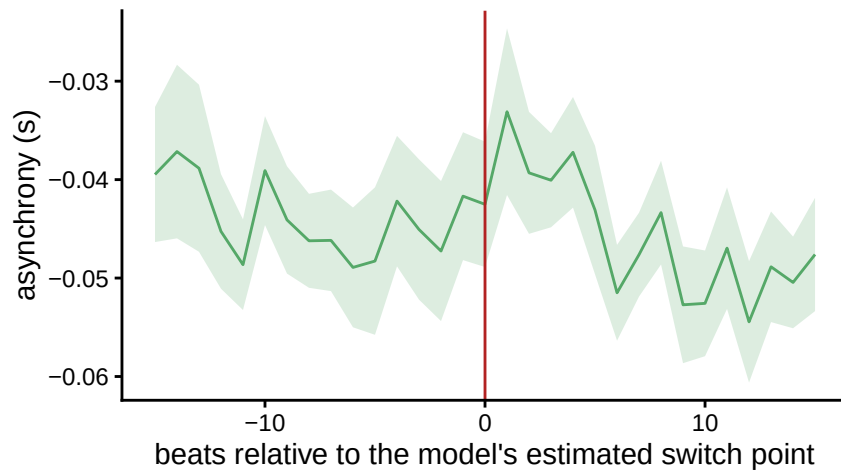


Aligned this way, the accent contrast is high before τ and drops sharply right around it – a useful independent check that the model isn't just fitting noise, using no model output except τ itself as an alignment point. Individual taps still look “zig-zaggy” by eye even after a real collapse (beat-to-beat noise is large relative to the shrunk accent), so a raw per-tap plot alone can look similar before and after a switch – this rolling, aligned-and-averaged statistic is what actually reveals the change. Where the model was unsure (the other 16 trials), it's for one of two reasons: the contrast never gets very high to begin with (nothing to switch away from), or it drops only partway rather than to zero (a weaker, less complete change) – both genuine data features, not a shortcoming of the fitting procedure.

Aligning spontaneous trials on the model's estimated switch

Confident spontaneous switches, aligned

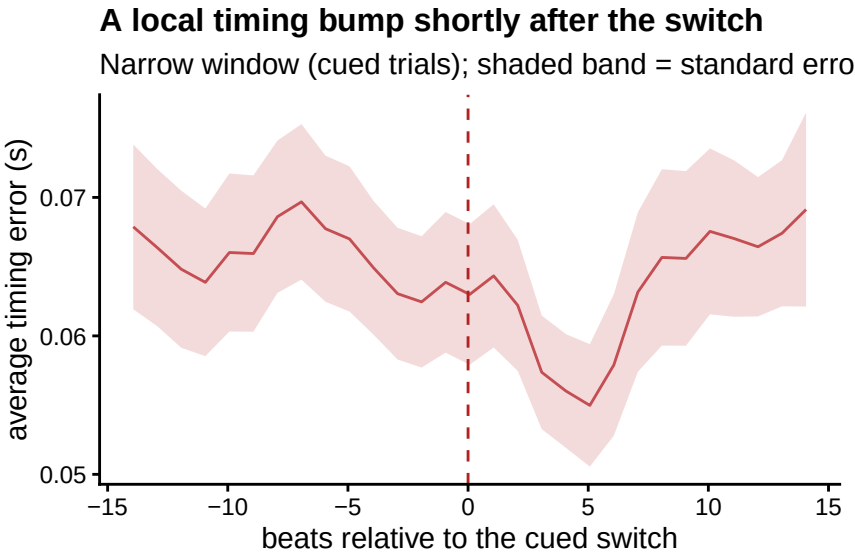
Mean $\pm$ SE across the 44 confident-switch trials



The main takeaway here is methodological: the model can pick out *which* spontaneous trials had a clear switch and roughly *when*, purely from the loudness/accent pattern, without ever being told a switch happened. Timing itself still doesn't show a strong signature at the switch, consistent with Part 1 – this alignment uses τ , it doesn't discover it.

A closer look: a small timing bump right after the switch

Timing wasn't reliable enough to detect a switch on its own (Part 1), but now that we have τ from the loudness model, it's worth looking at timing *around* that known point more closely than before. Using a narrow sliding window (much finer than earlier passes), there's a small, localised bump in timing error right after the switch: tap-to-click error dips exactly at the switch, then rises again for a few beats, before settling back down.



Now pooled across all 6 participants, the pattern is a modest dip in timing error a few beats *after* the switch (not right at it) before returning to baseline – present, but small relative to its own uncertainty band, so still worth treating as suggestive rather than a settled finding. If it holds up with more data, it would point to a brief timing wobble as the participant re-locks onto the new meter shortly after adopting its accent, distinct from the broader pre-vs-post trend.

Part 4: Comparing conditions and meters

Beyond τ itself, the model fits three other parameters per trial: s (how sharp the switch is), σ_{sw} (extra noise right at the switch), and σ_0 (baseline loudness noise, unrelated to the switch). Do any of these differ between cued and spontaneous trials, or between meter 2 and meter 3? Comparing posterior *means* alone can be misleading – it ignores how uncertain each trial's own estimate is. `02_fit_model.ipynb` computes a 94% credible interval on each group difference by resampling posterior draws across trials many times; with data from multiple subjects, that resampling has to respect that **subjects, not trials, are the independent unit** – 6 participants contributed 10-20 trials each, and trials from the same person are correlated, so treating every trial as an independent vote overstates precision. The method resamples within each subject first, then averages across subjects, not across trials directly. A difference only counts as credible if that interval excludes zero.

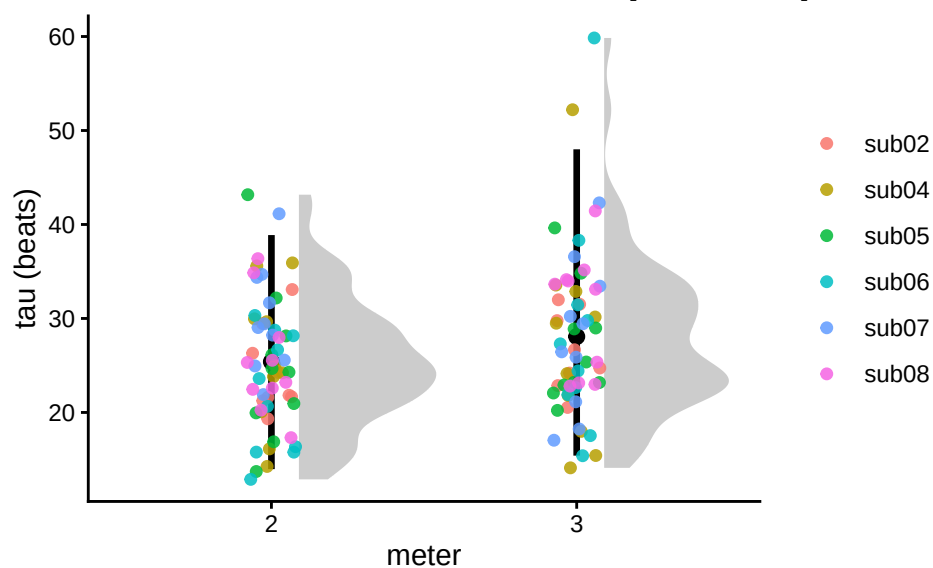
Comparison	Parameter	Group A mean	Group B mean	Difference	94% credible interval	Credible?
cued vs spontaneous	τ	29.351	24.039	5.312	[3.473, 7.082]	yes
cued vs spontaneous	s	0.176	0.175	0.001	[-0.024, 0.028]	no
cued vs spontaneous	σ_{sw}	0.450	0.446	0.003	[-0.1, 0.106]	no

Comparison	Parameter	Group A mean	Group B mean	Difference	94% credible interval	Credible?
cued vs spontaneous	sigma_0	0.747	0.825	-0.078	[-0.109, -0.047]	yes
meter 2 vs meter 3	tau	25.395	28.039	-2.644	[-4.518, -0.922]	yes
meter 2 vs meter 3	s	0.175	0.175	0.000	[-0.024, 0.026]	no
meter 2 vs meter 3	sigma_sw	0.451	0.444	0.007	[-0.099, 0.115]	no
meter 2 vs meter 3	sigma_0	0.770	0.800	-0.030	[-0.06, 0.001]	no

tau differs credibly between cued and spontaneous – not a new result, it’s the same reaction-lag effect already reported in Part 2. **s** and **sigma_sw** show no credible difference anywhere: switches are equally sharp and equally noisy at the switch, regardless of condition or meter, as far as this data can tell. **sigma_0** differs credibly between cued and spontaneous, but **not** between meters once subjects (not trials) are treated as the unit of replication – see below, this one looked credible under a naive analysis and doesn’t survive a proper one. The one finding that *is* credible by meter is **tau** itself. In the plot below: half-violin = distribution of per-trial means across all participants; coloured points = individual trials, jittered, by participant; black point + line = overall mean and 95% CI.

Switches complete later (slower) in meter-3 trials

meter 2 – meter 3 = -2.64 beats, 94% CI [-4.52, -0.92]



tau is credibly *lower* (earlier) in meter-2 trials than meter-3 – switches complete a few beats faster when starting from a 2-beat bar. This survives the subject-level analysis above, so it’s the one meter effect we’d actually stand behind. A speculative reading: a 2-beat bar (strong-weak) is a simpler pattern to abandon than a 3-beat one (strong-weak-weak), so committing to the new meter happens sooner.

What about sigma_0?

The meter effect on **sigma_0** (baseline noise) looked credible in an earlier pass of this analysis, using a resampling method that pooled all trials as if independent – reasonable-looking (large apparent sample, tight bars), but statistically thin for two separate reasons, both worth stating plainly rather than burying:

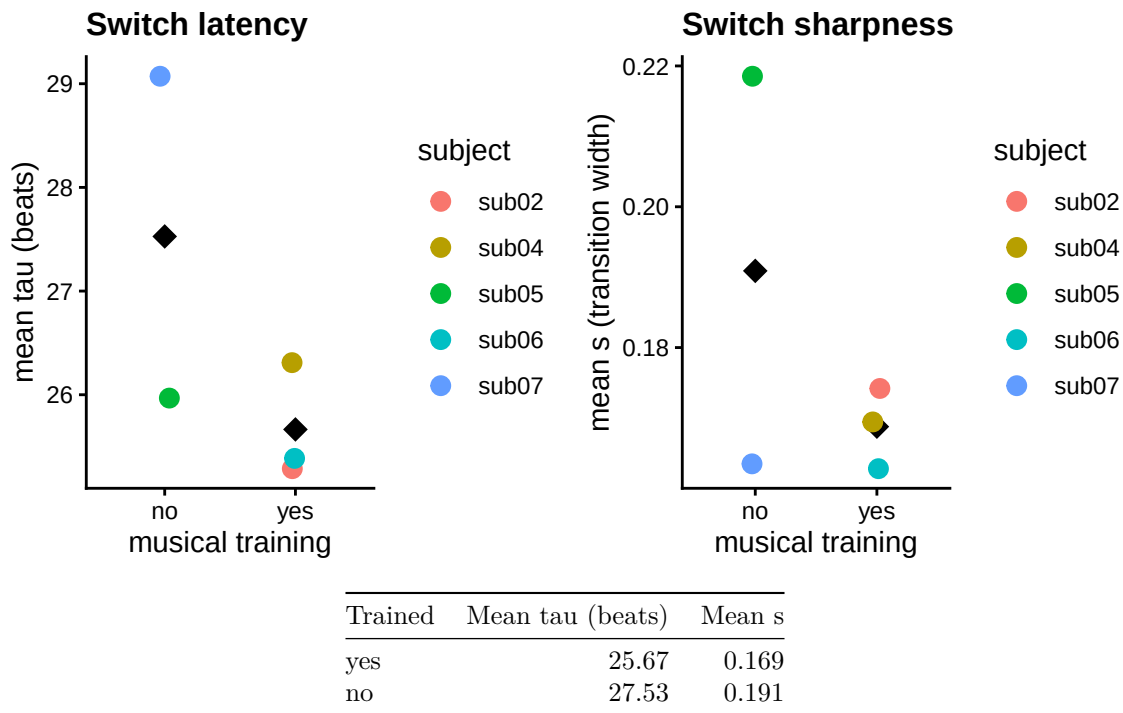
1. **Pseudo-replication.** With only 6 subjects contributing the trials, treating each trial as an independent unit lets subjects with more (or more extreme) trials dominate the estimate. Once subjects are correctly treated as the unit of replication (as in the **tau** result above), the meter effect on **sigma_0** is no longer credible (94% CI crosses zero).

2. **A structural confound.** Meter-3 trials have two “unaccented” bar positions to meter-2’s one. Checked directly in the raw data: unaccented taps are noisier than accented ones in both meters (SD ~9-11 dB vs. ~7 dB), so a meter with proportionally more unaccented positions will show higher *pooled* noise even with no true difference in participants’ consistency – the model fits one `sigma_0` per trial, not one per bar position, so it can’t separate “meter” from “proportion of unaccented beats.” These are perfectly confounded by the experiment’s own design (2-beat bars are structurally 50/50 accented, 3-beat bars are structurally 1/3-2/3).

Net: **no credible, uncounfounded evidence of a meter effect on baseline noise.** This replaces what an earlier, less careful pass of this analysis reported as a finding.

Does musical training predict switch behaviour?

A separate participant questionnaire (`demographics/music_training.xlsx`) records whether each participant reports musical training. Only 5 of the 6 fitted participants have a label (`sub08` isn’t in that file), split 3 trained / 2 untrained – **far too few to test anything** (`power_analysis.ipynb`’s Monte Carlo simulation, at ~20 trials/participant, estimates ~14-30 participants are needed for this exact question, depending on effect size). Shown here purely descriptively, not as a finding:



The group *means* (black diamonds above) do point the hypothesized way – trained participants switch a couple of beats earlier and somewhat more sharply than untrained ones. A pattern in the two panels together is worth calling out as suggestive: the two **trained** participants who overlap with the untrained group on one metric are coherent on *both* – trained participants cluster together on **tau** and **s** simultaneously. The two **untrained** participants instead split apart: one is comparably fast (**tau**) but noticeably less sharp (**s**) than the trained cluster, the other is comparably sharp but noticeably slower. That two people can each match the trained group on one axis while being the outlier on the other is itself weak evidence for a real, two-dimensional training effect, rather than one axis just being noisy – though with $n=2$ per “untrained” person, this is a pattern to flag for a bigger sample, not a claim. Worth being concrete about why this isn’t more than a hint: the “no” group is just sub05 and sub07, and sub07 is the same participant already flagged repeatedly in this report as our lowest-signal case (worst cued-validation confidence, worst spontaneous-switch confidence). A 2-person group where one member is a known outlier on data quality, not training, can

produce part of this pattern without training being the actual cause. Not dismissing it – just noting that $n=5$ can't distinguish “training effect” from “sub07 is noisy” here, which is exactly why the power analysis calls for ~15-30 participants before trusting this comparison.

Summary

Based on 6 participants (up from 1 in the previous version of this report).

- **Loudness, not timing, is where the switch signal lives** – tapping timing looks noisy and undifferentiated around a switch in every participant, and shows a consistent small anticipation effect (~25-70 ms early, a known tapping-literature phenomenon) unrelated to switching, so the model is built entirely on tap loudness.
- The switch-detection model was validated against cued trials (known instruction time): weaker pooled than it looked with one subject, and one participant (sub07) has almost no well-identified cued trials at all – restricted to trials the model itself is confident about (same wide-CI flag used for spontaneous trials), correlation with instruction time is 0.72 and lag is consistently positive (-1.7 to +22.3 beats). Real switches are close to instantaneous rather than gradual.
- For spontaneous trials, 44 of 60 trials show a confident, well-identified switch, but that varies a lot by participant (2/10 to 10/10) – confirmed independently by a model-free accent-contrast check with no fitting involved.
- **One credible group difference, pooled across participants:** τ (switch latency) is credibly faster in meter-2 trials than meter-3. A meter effect on σ_0 (baseline noise) looked credible under a naive analysis but doesn't survive once subjects (not trials) are treated as the independent unit, and is separately confounded with meter-3 having twice as many unaccented bar positions – dropped as a finding, kept in the report as a methodological note. Neither s nor σ_{sw} differ credibly by meter or condition.
- A first look at musical training ($n=5$, 3 trained/2 untrained) shows group means in the hypothesized direction, but one of only two “untrained” participants is sub07 – already flagged repeatedly as our lowest-signal case – so this can't be told apart from a data-quality artifact with this few people. The power analysis puts the real requirement at ~15-30 participants.

What works, what's next

What works:

1. The whole experiment pipeline works, end to end, across participants.
2. τ estimation works and finds real (credible) differences:
 - By meter: faster in meter 2, slower in meter 3 – survives the subject-level, confound-checked analysis, so this is the one we'd stand behind.
 - By condition: τ is later in cued than spontaneous, and σ_0 (baseline noise) is lower in cued than spontaneous – both expected/mechanical (cued trials have a known cue to react to, which structurally delays and steadies tapping) rather than new findings.
 - s and σ_{sw} show no credible difference anywhere (meter or condition).
3. Musical training shows an effect, but it's anecdotal only ($n=5$).

Future work:

1. Better recording of tapping (reduce noise/dropouts seen in some participants).
2. More trials per participant.
3. A hierarchical model, pooling properly across trials and participants.
4. A better measure of musical training (a proper questionnaire, not yes/no).